Distribute System : 03/10/2025

Discuss in detail Synchronization in distributed system.

HB Synchronization in distributed system refers to the way different machines or processes are kept in agreement when working together which helps ensure consistency, order and co-ordination across all the nodes.

It's Importance

- It ensures data consistency across all nodes.
- Maintains correct order of events and operations
- prevents conflicts when accessing shared resources
- Improves system reliability and accuracy.

Types of synchronization

a) - Clock Synchronization :-
It ensures all machines are working with similar time.
e.g. Network time protocol (NTP) & Berkeley Algorithm.

b) Process Synchronization
- Helps control how multiple processes access shared resources like files & variables.
- Prevents two ~~variables~~ processes from changing the same data at the same time

Techniques used include locks, semaphores & token based methods

c Data Synchronization
Ensures that data stored in different locations is consistent and up to date
Especially important when systems use backups or replicate data across servers

Challenges of Synchronization

- Network delays: Messages between Machines can arrive late or out of order
- Clock drift: Clocks on different Machines can become unsynchronized
- Concurrency issue: when multiple processes act at the same time, it's hard to manage who gets access first
- Failure Handling: If one node crashes, the system must still stay synchronized